The tangent groupoid, its zoom, and whole trajectories

Correction, 2026-10-27. The earlier numbered summary below and corresponding passages are historical and their broad identifications are withdrawn. The groupoid zoom is a geometric rescaling; identifying it with integration of fluctuations or with a running physical action constant requires additional arguments. A real-time Hamilton principal function composes by stationary elimination on an appropriate branch, not by an unconditional global minimum. Euclidean Brownian scaling is not a probability law for real-time Feynman paths, and an unqualified claim that quantum paths have Hausdorff dimension two is not justified. The precise state-level construction, the different scales and a Gaussian scaling test are given in the later tangent-groupoid formulation. The constant-force calculation remains a separately scoped example.

Result, 2026-09-27 (user's question: what Connes's tangent groupoid adds, beyond the user's 1998 and 2003 readings, and whether it extends to a whole trajectory). Three statements.

1. **The zoom is the renormalization.** Debord and Skandalis's action of $\mathbb{R}_+^*$ on the tangent groupoid, $(x, y, \varepsilon) \mapsto (x, y, \lambda^{-1}\varepsilon)$ and $(x, v, 0) \mapsto (x, \lambda v, 0)$, is the dilation family of the fifth-postulate note (§4, Theorem D) and the control transformation $h \mapsto \mu h$ of the user's 1998 note. By van Erp and Yuncken, pseudodifferential operators are exactly the distributions on the tangent groupoid that are homogeneous under the zoom up to smoothing terms; their restriction to $\varepsilon = 0$ is the principal symbol. What survives the passage $\varepsilon \to 0$ is therefore the principal symbol, a function of place and velocity together.
2. **Trajectories live on a time-graded groupoid.** The tangent groupoid composes chords at one fixed ε, and at $\varepsilon = 0$ it composes only vectors at one point. Refining a trajectory into more pieces is an operation on the groupoid $(x, y, u) \circ (y, z, v) = (x, z, u + v)$ that the user introduced in 2003 as the "elementary school" groupoid. Refinement consistency is a morphism property on it: classically $\min_y[S(x, y, r) + S(y, z, u - r)] = S(x, z, u)$, satisfied by Hamilton's principal function, the exact discrete Lagrangian of discrete mechanics, and for Galileo's constant force by the corrected action ℓ_h of the refinement note, eq. (2) (Proposition 1); quantum mechanically the convolution law $K(u+v) = K(u) * K(v)$. The first is the $\hbar \to 0$ tropical shadow of the second (Maslov dequantization).
3. **Along a quantum trajectory the ultimate velocity fails.** Quantum paths have Hausdorff dimension two: $\langle \Delta x^2 \rangle \sim \hbar\tau/M$ over a step τ, so $\Delta x/\tau$ diverges like $\tau^{-1/2}$. The tangent groupoid's $\varepsilon \to 0$ fibre, where $(y - x)/\varepsilon \to v$, is reached by ballistic scaling; quantum trajectories need the diffusive scaling $\Delta x^2 \propto \tau$, the user's 1998 remark that the path groupoid relates to Connes's only after $(x, y, \varepsilon) \mapsto (x, y, \varepsilon^2)$, and his 2003 remark on the $\sqrt{t}$ of Itô's calculus. Newton's *velocitas ultima* is the ballistic boundary fibre, which quantum paths do not reach.

The constructions are established; the identifications are readings, and Proposition 1 is elementary. The note gives the atlas's Newton row a geometric home: one step is an arrow of the tangent groupoid, a trajectory is a factorization in the time-graded groupoid, the zoom is the renormalization, and the principal symbol is what survives.

1. Connes's construction

For a manifold M, the tangent groupoid is $\mathbb{G}_M = (M \times M \times (0, 1]) \cup (TM \times \{0\})$, with the pair-groupoid law at each $\varepsilon > 0$, the fibrewise vector addition $(x, X) \circ (x, Y) = (x, X + Y)$ at $\varepsilon = 0$, and the topology in which $(x_n, y_n, \varepsilon_n) \to (x, v, 0)$ iff $x_n \to x$ and $(y_n - x_n)/\varepsilon_n \to v$ (in a chart). Its C^*-algebra is a continuous field whose fibre at $\varepsilon > 0$ is the compact operators on $L^2(M)$ and at $\varepsilon = 0$ is $C_0(T^*M)$ by Fourier transform; Connes used the deformation to prove the index theorem (A. Connes, *Noncommutative Geometry*, Academic Press 1994, §II.5, metadata),

and the construction generalizes the Moyal rule (Cariñena, Clemente-Gallardo, Follana, Gracia-Bondía, Rivero and Várilly 1999, abstract; Landsman 2002, arXiv:math-ph/0208004, abstract). The $\varepsilon = 0$ fibre is where a chord has become a velocity at a place: Newton's *velocitas ultima*, "certain and definite", with place and velocity commuting. The fibre at $\varepsilon > 0$ is noncommutative.

2. The zoom action and what survives

Debord and Skandalis (2014) (metadata) define the action of $\mathbb{R}_+^*$ on the adiabatic (tangent) groupoid displayed in the summary and identify the crossed product with an ideal of pseudodifferential operators; van Erp and Yuncken (2019) (metadata) characterize pseudodifferential operators as the distributions on the tangent groupoid, properly supported and transversal, that are homogeneous under the zoom modulo smooth ones. Three identifications follow (readings):

- the zoom on the $\varepsilon = 0$ fibre, $v \mapsto \lambda v$, is dual on T^*M to the scaling that carries the Moyal product with constant $\hbar$ to the one with constant $\lambda^{\pm 2}\hbar$ (fifth-postulate note, §4);
- the user's 1998 "dressed series", invariant under $h \mapsto \mu h$ (arXiv:quant-ph/9803035, eq. (10)), is a zoom-invariance condition, and his Ehrenfest remark (eq. (11)) is the statement that the equations of motion are zoom-invariant (Theorem A of the fifth-postulate note);
- the principal symbol, the $\varepsilon = 0$ restriction of a zoom-homogeneous distribution, is the classical datum that survives the limit. The atlas's question "what survives refinement" has here a precise answer for one step.

3. Whole trajectories: the time-graded groupoid

The user's 2003 note (Rivero, "Flashes of noncommutativity", arXiv:math/0302285, full read) introduces the groupoid over configuration space with arrows (x, y, u), u an elapsed time, and law $(x, y, u) \circ (y, z, v) = (x, z, u + v)$. Its convolution algebra has product $(AB)(x, z, t) = \iint A(x, y, r)\, B(y, z, t - r)\, dy\, dr$; Fourier transformation in t gives, with $\varepsilon \sim 1/\hat{t}$, the product of the $\varepsilon > 0$ part of the tangent groupoid, so the latter's algebra sits inside the former's (the 2003 note's second flash). Refining a trajectory is factoring an arrow (x, z, u) through intermediate points and times, which is Newton's insertion of $t_{2.6}$.

Proposition 1. (a) A function $S(x, y, u)$ satisfies $\min_y [S(x, y, r) + S(y, z, u - r)] = S(x, z, u)$ for all $0 < r < u$ exactly when it is a morphism of the time-graded groupoid into the $(\min, +)$ semiring. For Galileo's constant force F on a mass M this holds for

$$S(x, y, u) = \frac{M(y - x)^2}{2u} + \frac{Fu}{2}(x + y) - \frac{F^2 u^3}{24M},$$

the corrected action ℓ_u of the refinement note, eq. (2), which is the action along the true path. (b) A kernel $K(x, y, u)$ satisfies $\int K(x, y, r) K(y, z, u - r)\, dy = K(x, z, u)$ exactly when it is a morphism into the convolution algebra; the constant-force propagator U_u of the refinement note, eq. (3), does.

Proof. (a) and (b) restate the morphism property. The displayed S is the classical action of the path $q(s) = x + (y - x)s/u + \frac{F}{2M} s(s - u)$, which is the refinement note's ℓ_u; its composition law is eq. (2) there, and U_u's is eq. (3). □

The displayed S is the exact discrete Lagrangian of discrete mechanics (Marsden and West 2001, metadata), and discrete Lagrangian mechanics on groupoids is Weinstein's (1996) (metadata). The kick–drift–kick cell $S_h = K_{Fh/2} D_h K_{Fh/2}$ of the refinement note is the variational (Störmer–Verlet) integrator of the trapezoidal discrete Lagrangian L_h, and the cubic counterterm turns L_h into the exact one. The two laws of Proposition 1 are related by Maslov dequantization: in Euclidean form, $-\hbar \log \int e^{-S/\hbar} \to \min S$ as $\hbar \to 0$ (Litvinov 2005, metadata). So refinement consistency of a whole trajectory is the morphism property, tropical in classical mechanics and convolutional in quantum mechanics.

4. Quantum trajectories miss the ultimate velocity

In a path integral the typical increment over a step τ has $\langle \Delta x^2 \rangle \sim \hbar\tau/M$, so the paths have Hausdorff dimension two (Abbott and Wise 1981, metadata), and the difference quotient $\Delta x/\tau \sim (\hbar/M\tau)^{1/2}$ diverges. The tangent groupoid reaches its $\varepsilon = 0$ fibre along ballistic sequences, $(y - x)/\varepsilon \to v$. Quantum trajectories approach along diffusive ones, $\Delta x^2 \propto \tau$: the user's 1998 remark that Connes's groupoid relates to the groupoid of paths only after the rescaling $(x, y, \varepsilon) \mapsto (x, y, \varepsilon^2)$, and his 2003 remark on the $\sqrt{t}$ of Itô's calculus, are this fact. The dimensionless ratio $M\,\Delta x^2/(\hbar\tau)$ is the action per step in units of $\hbar$: ballistic refinement sends it to zero (the classical limit, Newton's ultimate velocity), diffusive refinement holds it fixed (the dimension ladder: refining makes each Newtonian cell quantum). The failure of joint determinacy of the fifth-postulate note is, along a trajectory, the failure of the ballistic limit.

5. Consequence for STATE

The atlas's Newton row gains a geometric home: one step is an arrow of the tangent groupoid, a trajectory is a factorization in the time-graded groupoid, refinement consistency is the morphism property of Proposition 1, the zoom action is the renormalization, and the principal symbol is what survives $\varepsilon \to 0$. A natural next question for the gauge atlas is the corresponding groupoid of the lattice: the holonomy groupoid for two dimensions, where series moves are compositions, and a higher (double) groupoid for the parallel moves of three and four dimensions.